

module4

Gafur Mammadov

2025-03-20

```
colleges2015_clean = drop_na(Colleges2015)
colleges2015_clean = colleges2015_clean %>% select(-GraduationRate4Yr, -instnm)
colleges2015_clean = colleges2015_clean[-1, ]
head(colleges2015_clean)
```

```
##   GraduationRate6Yr PercentFinancialAid PercentStudentLoans
## 2                68                94                59
## 3                27                88                75
## 4                48               100                63
## 5                48               100                81
## 6                71               100                67
## 7                49               100                84
##   TotalPriceInStateOnCampus TotalPriceOutStateOnCampus
## 2                   51708                   51708
## 3                   22896                   30666
## 4                   31010                   31010
## 5                   45089                   45089
## 6                   52085                   52085
## 7                   36511                   36511
##   TotalPriceInStateWithFamily PercentAdmitted TotalEnrollment Applicants
## 2                   42508                72                7256        9367
## 3                   14756                65                5627        7901
## 4                   24910                39                 522         699
## 5                   36379                66                1482        1421
## 6                   41265                80                1376        2703
## 7                   29389                54                1154        4206
##   Enrolled Admitted SatMath75th SatReading75th SatWriting75th StudToFacRatio
## 2      868    6762         620         600         620             11
## 3     1461    5166         480         470         457             19
## 4       49     272         545         570         520              8
## 5      188     936         650         620         610              9
## 6      456    2157         560         600         580             12
## 7      333    2258         530         510         490             17
```

```
set.seed(123)
n <- nrow(colleges2015_clean)
train_indices <- sample(1:n, 400)
train_data <- colleges2015_clean[train_indices, ]
test_data <- colleges2015_clean[-train_indices, ]

head(train_data)
```

```
##      GraduationRate6Yr PercentFinancialAid PercentStudentLoans
## 416          67          98          51
## 464          77          81          28
## 180          53          94          54
## 527          66          99          68
## 196           5          66           0
## 119          55         100          45
##      TotalPriceInStateOnCampus TotalPriceOutStateOnCampus
## 416          52577          52577
## 464          58270          58270
## 180          26214          40782
## 527          56111          56111
## 196          37930          37930
## 119          35380          35380
##      TotalPriceInStateWithFamily PercentAdmitted TotalEnrollment Applicants
## 416          38900          70          11861          9841
## 464          46470          43          10324          18423
## 180          14584          57          32058          13970
## 527          45094          47          8334          8179
## 196          26830          97          526          403
## 119          33950          49          1550          695
##      Enrolled Admitted SatMath75th SatReading75th SatWriting75th StudToFacRatio
## 416      2048      6880          580          600          590          20
## 464      2078      7974          650          630          640          13
## 180      3447      7914          590          580          570          21
## 527       724      3859          570          560          560          17
## 196       351       390          570          600          570          20
## 119       200       341          600          480          430          16
```

Question1

```
formula = as.formula(paste("GraduationRate6Yr ~", paste(names(train_data)[2:15], collapse= "+")))
library(leaps)
colleges_regfit=regsubsets(formula, data=train_data, nvmax = NULL)
reg_summary = summary(colleges_regfit)
summary(colleges_regfit)
```

```
## Subset selection object
## Call: regsubsets.formula(formula, data = train_data, nvmax = NULL)
## 14 Variables (and intercept)
##
```

	Forced in	Forced out
## PercentFinancialAid	FALSE	FALSE
## PercentStudentLoans	FALSE	FALSE
## TotalPriceInStateOnCampus	FALSE	FALSE
## TotalPriceOutStateOnCampus	FALSE	FALSE
## TotalPriceInStateWithFamily	FALSE	FALSE
## PercentAdmitted	FALSE	FALSE
## TotalEnrollment	FALSE	FALSE
## Applicants	FALSE	FALSE
## Enrolled	FALSE	FALSE
## Admitted	FALSE	FALSE
## SatMath75th	FALSE	FALSE
## SatReading75th	FALSE	FALSE
## SatWriting75th	FALSE	FALSE
## StudToFacRatio	FALSE	FALSE

```
## 1 subsets of each size up to 14
## Selection Algorithm: exhaustive
##
```

	PercentFinancialAid	PercentStudentLoans	TotalPriceInStateOnCampus
## 1 (1)	" "	" "	" "
## 2 (1)	" "	" "	" "
## 3 (1)	" "	" "	" "
## 4 (1)	" "	" "	" "
## 5 (1)	" "	" "	" "
## 6 (1)	" "	"*"	" "
## 7 (1)	" "	"*"	" "
## 8 (1)	" "	"*"	" "
## 9 (1)	" "	"*"	" "
## 10 (1)	" "	"*"	" "
## 11 (1)	" "	"*"	"*"
## 12 (1)	" "	"*"	"*"
## 13 (1)	" "	"*"	"*"
## 14 (1)	"*"	"*"	"*"

```
##
```

	TotalPriceOutStateOnCampus	TotalPriceInStateWithFamily
## 1 (1)	" "	" "
## 2 (1)	"*"	" "
## 3 (1)	"*"	" "
## 4 (1)	"*"	" "
## 5 (1)	"*"	" "
## 6 (1)	"*"	" "
## 7 (1)	"*"	" "
## 8 (1)	"*"	" "
## 9 (1)	"*"	" "
## 10 (1)	"*"	" "
## 11 (1)	"*"	"*"
## 12 (1)	"*"	"*"
## 13 (1)	"*"	"*"
## 14 (1)	"*"	"*"

```
##
```

	PercentAdmitted	TotalEnrollment	Applicants	Enrolled	Admitted
## 1 (1)	" "	" "	" "	" "	" "

```
## 2 ( 1 ) " " " " " "
## 3 ( 1 ) " " " " " "
## 4 ( 1 ) " " "*" " " "
## 5 ( 1 ) " " "*" " " "
## 6 ( 1 ) " " "*" " " "
## 7 ( 1 ) " " "*" " " "
## 8 ( 1 ) "*" "*" " " "*" " "
## 9 ( 1 ) "*" "*" " " "*" " "
## 10 ( 1 ) "*" "*" "*" "*" " "
## 11 ( 1 ) "*" "*" "*" "*" " "
## 12 ( 1 ) "*" "*" "*" "*" "*"
## 13 ( 1 ) "*" "*" "*" "*" "*"
## 14 ( 1 ) "*" "*" "*" "*" "*"
##          SatMath75th SatReading75th SatWriting75th StudToFacRatio
## 1 ( 1 ) "*" " " " " " "
## 2 ( 1 ) "*" " " " " " "
## 3 ( 1 ) "*" "*" " " " "
## 4 ( 1 ) "*" " " " " " "
## 5 ( 1 ) "*" "*" " " " "
## 6 ( 1 ) "*" "*" " " " "
## 7 ( 1 ) "*" "*" "*" " "
## 8 ( 1 ) "*" "*" "*" " "
## 9 ( 1 ) "*" "*" "*" "*"
## 10 ( 1 ) "*" "*" "*" "*"
## 11 ( 1 ) "*" "*" "*" " "
## 12 ( 1 ) "*" "*" "*" " "
## 13 ( 1 ) "*" "*" "*" "*"
## 14 ( 1 ) "*" "*" "*" "*"

```

```
best_size <- which.min(reg_summary$bic)
selected_vars <- names(coef(colleges_regfit, best_size))[-1] # remove intercept

formula_selected <- as.formula(paste("GraduationRate6Yr ~", paste(selected_vars, collapse = "+")))
final_lm <- lm(formula_selected, data = train_data)

pred_regfit <- predict(final_lm, newdata = test_data)

sqrt(mean((test_data$GraduationRate6Yr - pred_regfit)^2))

```

```
## [1] 10.01723

```

```
selected_vars <- names(coef(colleges_regfit, best_size))[-1] # remove intercept
print(selected_vars)

```

```
## [1] "TotalPriceOutStateOnCampus" "TotalEnrollment"
## [3] "Enrolled" "SatMath75th"
## [5] "SatReading75th"

```

Question 2

```
library(caret)
```

```
## Loading required package: lattice
```

```
##  
## Attaching package: 'caret'
```

```
## The following object is masked from 'package:purrr':  
##  
## lift
```

```
train_data$GraduationRate6Yr <- as.numeric(train_data$GraduationRate6Yr)  
test_data$GraduationRate6Yr <- as.numeric(test_data$GraduationRate6Yr)
```

```
college_train_lm = train(  
  GraduationRate6Yr ~ .,  
  data = train_data,  
  method = "lm",  
  trControl = trainControl(  
    method = "cv",  
    number = 10  
  )  
)
```

```
pred_lm <- predict(college_train_lm, newdata = test_data)
```

```
library(Metrics)
```

```
##  
## Attaching package: 'Metrics'
```

```
## The following objects are masked from 'package:caret':  
##  
## precision, recall
```

```
rmse(test_data$GraduationRate6Yr, pred_lm)
```

```
## [1] 9.875575
```

Question 3

```
library(glmnet)
```

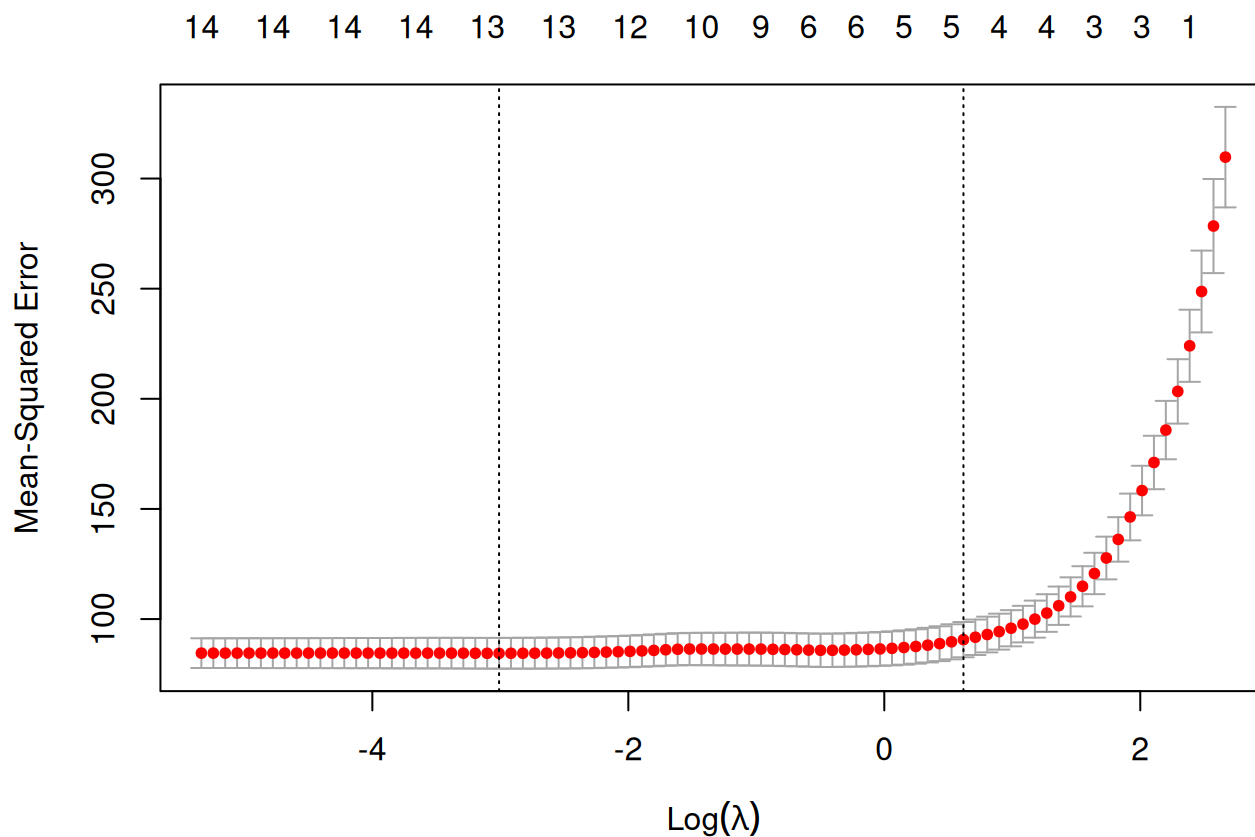
```
## Loading required package: Matrix
```

```
##  
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':  
##  
## expand, pack, unpack
```

```
## Loaded glmnet 4.1-8
```

```
y = as.numeric(train_data$GraduationRate6Yr)  
formula = as.formula(paste("GraduationRate6Yr ~ ", paste(names(train_data)[2:15], collapse = "+")))  
  
X = model.matrix(formula, data=train_data)[, -1]  
  
cv.ridge=cv.glmnet(X, y, alpha=1, nfolds = 10)  
plot(cv.ridge)
```



```
ridge.opt = glmnet(X, y, lambda = cv.ridge$lambda.min, alpha=1)
coef(ridge.opt)
```

```
## 15 x 1 sparse Matrix of class "dgCMatrix"
##                               s0
## (Intercept)                -5.441647e+01
## PercentFinancialAid         -9.332133e-03
## PercentStudentLoans          7.377832e-02
## TotalPriceInStateOnCampus    .
## TotalPriceOutStateOnCampus   3.531512e-04
## TotalPriceInStateWithFamily  6.012070e-05
## PercentAdmitted              -3.586795e-02
## TotalEnrollment             -3.203768e-04
## Applicants                   -9.033906e-05
## Enrolled                     3.589529e-03
## Admitted                     2.573800e-04
## SatMath75th                  9.465302e-02
## SatReading75th               3.817199e-02
## SatWriting75th               2.922905e-02
## StudToFacRatio               -1.547089e-01
```

```
# Create X matrix for test data
X_test = model.matrix(formula, data = test_data)[,-1]

# Predict on test set using ridge
ridge_preds = predict(ridge.opt, s = cv.ridge$lambda.min, newx = X_test)

# Calculate RMSE
rmse(test_data$GraduationRate6Yr, ridge_preds)
```

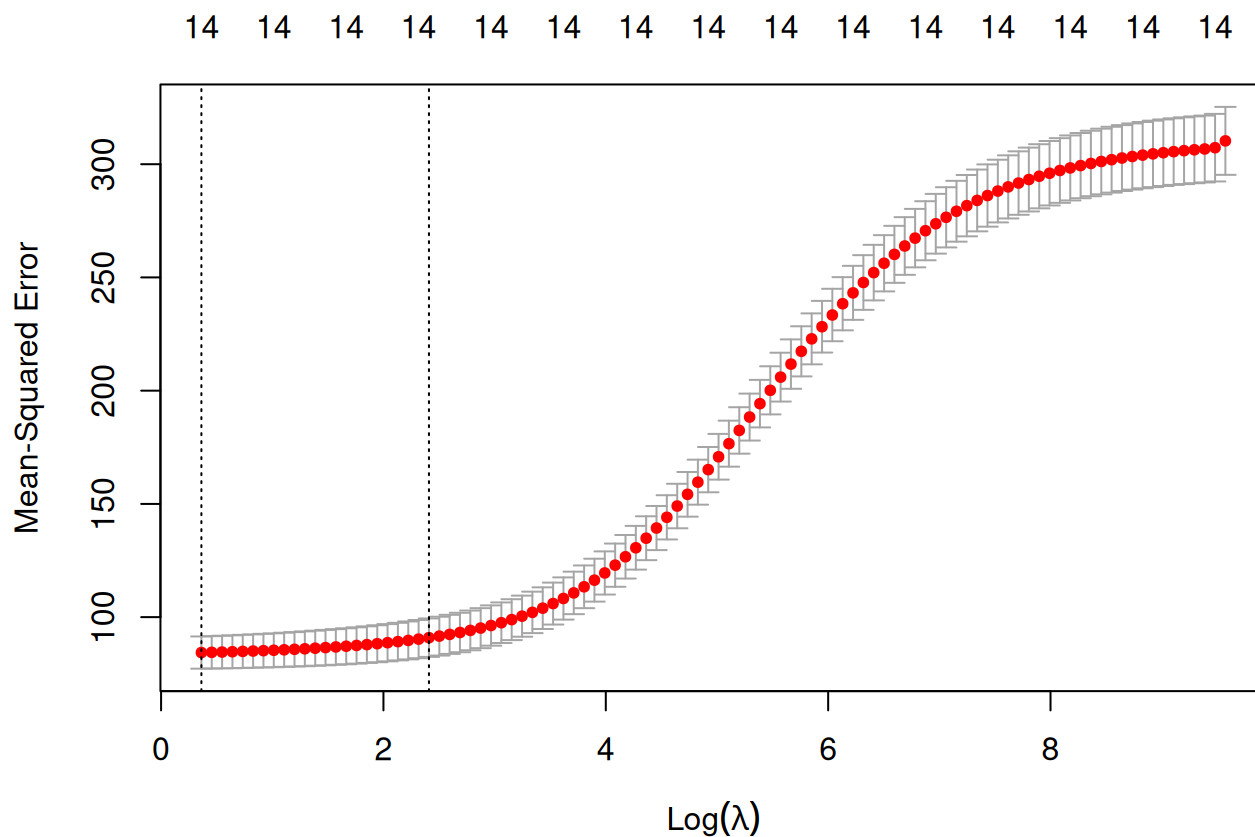
```
## [1] 9.872572
```

Question 4

```
library(glmnet)
y = as.numeric(train_data$GraduationRate6Yr)
formula = as.formula(paste("GraduationRate6Yr ~ ", paste(names(train_data)[2:15], collapse= "+")))

X = model.matrix(formula,data=train_data)[,-1]

cv.ridge=cv.glmnet(X, y, alpha=0, nfolds = 10)
plot(cv.ridge)
```



```
ridge.opt = glmnet(X, y, lambda = cv.ridge$lambda.min, alpha=0)
coef(ridge.opt)
```

```
## 15 x 1 sparse Matrix of class "dgCMatrix"
##                               s0
## (Intercept)                  -4.595156e+01
## PercentFinancialAid           -2.983763e-02
## PercentStudentLoans           5.916743e-02
## TotalPriceInStateOnCampus      2.289540e-05
## TotalPriceOutStateOnCampus     2.885133e-04
## TotalPriceInStateWithFamily    8.282102e-05
## PercentAdmitted               -2.833735e-02
## TotalEnrollment               -1.052174e-04
## Applicants                    -5.649273e-05
## Enrolled                      1.679374e-03
## Admitted                      3.793573e-04
## SatMath75th                   7.109959e-02
## SatReading75th                4.474890e-02
## SatWriting75th                3.762797e-02
## StudToFacRatio                -1.559199e-01
```



```
# Create X matrix for test data
X_test = model.matrix(formula, data = test_data)[,-1]

# Predict on test set using ridge
ridge_preds = predict(ridge.opt, s = cv.ridge$lambda.min, newx = X_test)

# Calculate RMSE
rmse(test_data$GraduationRate6Yr, ridge_preds)
```

```
## [1] 9.765799
```

Based on the RMSE results, Ridge regression has the lowest RMSE with 9.765799. The Larger the coefficient, more important that variable becomes. That's why, student-to-faculty ratio is the most important.